Mapping Natural-Disaster Damage from Satellite Imagery

Cross-disaster transfer on xBD: pretrained versus from-scratch change-detection CNNs

Artur Zavistovskyi, Bhuvanesh Dinesh Wadhwani, Anastasiia Khitrova, Yitian Zhou
Deep Learning Final Project

Abstract. This project addresses two questions about automated building-damage classification on xBD: whether performance transfers to a disaster type withheld entirely from training, and whether ImageNet pretraining improves that transfer over a comparable encoder trained from scratch. Two siamese change-detection CNNs, differing only in the encoder (an ImageNet-pretrained ResNet-18 versus a compact from-scratch encoder), are trained on 14 non-fire disaster events under identical patches, splits and class weights, then evaluated in-distribution and zero-shot on five held-out wildfire events. The pretrained model reaches 0.808 accuracy and 0.650 macro-F1 in distribution, falling to 0.774 accuracy and 0.429 macro-F1 on wildfire, a 33.9% relative macro-F1 loss. This aggregate drop is not a general representation failure: destroyed scores higher out of domain than in it (F1 0.758 versus 0.687), no-damage precision reaches 0.966, and ROC-AUC holds at 0.814. The degradation is confined to the two intermediate classes, 19.1% of training patches but only 1.7% of wildfire patches, so the learned features transfer while the decision calibration, tuned to the training-domain class prior, does not. Pretraining outperforms training from scratch on every split and metric, most where it matters: +0.143 accuracy and +0.052 macro-F1 on wildfire. A third analysis joins 185,045 labelled buildings to census block groups and finds an apparent income-damage association ($\beta = +0.225$, $p = 0.0009$) that disappears once event fixed effects are added ($\beta = +0.057$, $p = 0.44$): the confound is which disaster struck which place, not income. Recovery capacity tells a different story: priced in dollars against HAZUS ratios, validated against NFIP, CAL FIRE and FEMA records, 60% of the poorest income quartile's flood loss is covered by neither insurance nor federal aid against 33% in the richest, and auditing the Part 2 models' own predictions shows an automated relief rule would misdirect $30–78M of every $1B on seen disaster types and $112–200M on wildfire, the type never seen in training.

1 Introduction

Automated damage maps are operationally useful only if they generalise to disaster types with no labelled data yet. In-distribution accuracy does not measure this: a model can perform well on held-out data from disaster types seen in training while failing on one it has never encountered. This report addresses two questions: how much performance transfers to a disaster type withheld entirely from training, and whether ImageNet pretraining improves that transfer over a comparable from-scratch encoder, since pretraining costs nothing extra in labelling.

We study these questions on xBD [1], the dataset behind the xView2 challenge [9], which pairs pre- and post-event satellite tiles with building footprints and a four-level damage scale. Prior work fuses the two time steps inside the network [2]; we instead keep the comparison explicit in the classification head (Section 5.2). The task is change detection at building level: given co-registered pre- and post-event crops of a structure, predict no-damage, minor-damage, major-damage or destroyed.

The experimental design avoids a random split: five wildfire events are withheld entirely as a zero-shot test set, while the official xView2 evaluation directories give an in-distribution reference from the same trained model. Both encoders are evaluated identically on both sets, so the results address transfer and pretraining together rather than in isolation.

2 Data

2.1 Source and access

We use the full xBD release: 11,034 scenes over 19 events, each contributing a 1024 × 1024 RGB pre-disaster tile, the matching post-disaster tile, and a GeoJSON label file. That is 22,068 images and 425,368 annotated building polygons on the post-disaster side, obtained through a public mirror of the xView2 challenge release.[1]

Two properties of the source were verified rather than assumed. First, the tiles are co-registered: pixel (x, y) denotes the same ground location before and after the event, so cropping an identical bounding box from both is valid; otherwise the siamese design would compare different places. Second, the format is uniform: all 788 files in a stratified quality-control sample decode as 1024 × 1024 RGB PNG with no read errors, so no resolution standardisation or band alignment is needed. Damage labels sit only in the post-disaster GeoJSON, under `features.xy[].properties.subtype`; the same file carries `features.lng_lat`, giving per-building WGS84 coordinates, so Part 3's geo-referencing is already present.

2.2 Split design

The split is scene-level and was produced by a dedicated indexing script rather than inside a training notebook, so every model in the project trains on an identical partition (Table 1).

Two aspects merit comment. First, `test_id` is drawn from the official `test` and `hold` directories, never part of any training pool and, unlike `val`, never used for tuning, so it gives an honest headline number. Second, assignment happens at scene level, guaranteeing the pre- and post-disaster image of a location cannot separate and no building appears in two splits.

The five held-out events are `pinery-bushfire`, `portugal-wildfire`, `santa-rosa-wildfire`, `socal-fire` and `woolsey-fire`. An automated check confirms zero wildfire scenes in `train` or `val`, and zero

[1] Kaggle dataset `qianlanzz/xbd-dataset`, layout `xbd/{hold,test,tier1,tier3}/{images,labels}`.

Table 1: Split design and sizes. Only `train`/`val` depend on the random seed; both test sets are defined by the data itself.

Split	Rule	Scenes	Polygons
`train`	tier1+3, non-fire	3,135	225,506
`val`	tier1+3, non-fire	392	30,794
`test_id`	test+hold, non-fire	1,135	91,755
`test_ood`	wildfire, all	6,372	77,313
Total		11,034	425,368

event overlap between the training pool and `test_ood`.

3 Exploratory analysis

3.1 Class imbalance is severe and split-dependent

no-damage accounts for 72.0% of training polygons and the rarest class is 11.6× smaller (Fig. 1). More importantly the composition is not stable across splits: `test_ood` is 80.2 / 0.9 / 0.9 / 14.7% against 72.0 / 9.7 / 8.4 / 6.2% in `train`.

Plain accuracy is therefore a poor decision metric: always predicting no-damage scores 0.72 while flagging no damage at all. We report macro-F1 and per-class precision/recall instead, use inverse-frequency class weights from `train` only (0.330/2.379/2.910/4.919), and key checkpointing and early stopping on validation macro-F1. Weighting was preferred to resampling: oversampling would repeat the same destroyed buildings each epoch and invite memorisation, while undersampling would discard most of the majority signal.

3.2 Wildfire damage is bimodal

Averaged over events, the non-fire domain is 70.2 / 9.9 / 7.2 / 6.5% across the four classes while the wildfire domain is 78.8 / 1.2 / 1.1 / 14.8%. Fire either leaves a building intact or destroys it; intermediate states barely occur. Figure 2 shows this per event, and also how much the non-fire events differ among themselves: `mexico-earthquake` is 99% undamaged, `hurricane-matthew` is 51% minor-damage, `hurricane-harvey` is 35% major-damage.

This bimodality operationalises the project's central hypothesis: a model that learns the non-fire class prior will be miscalibrated on fire even if its visual features transfer perfectly. It also predicts what to look for in the out-of-domain errors: a prior shift should produce a consistent direction, whereas a representation failure should produce diffuse, undirected errors.

3.3 Scene occupancy and patch size

Empty scenes are common and strongly event-dependent (Fig. 3): 9.8% of training scenes contain no buildings, against 54.1% of wildfire scenes, whose median building count is zero, because the tier3 fire events cover large stretches of forest and scrubland. An empty scene yields no samples but still costs a full 1024 × 1024 decode, so we sample only scenes with at least five usable buildings. This is a sampling decision, not a cleaning one: nothing is deleted, and the identical rule applies to all four splits so it cannot tilt the comparison.

Buildings are small in absolute pixels: median longest bounding-box side 33.9 px, 90th percentile 60.9 px, with a thin tail to 299 px at the 99th. A 64×64 patch covers just over 90% of footprints near native resolution; larger mostly upsamples empty ground at quadratic cost, smaller destroys the roof texture separating minor from major. We keep a 10 px margin, leaving a 44×44 px window for the building. Damage is often more visible around a building than on it (debris, scorch marks, standing water), but the margin stays modest deliberately, for reasons Section 7 makes concrete.

3.4 A confound in the acquisition geometry

Ground sample distance and off-nadir angle are similar across the two domains (2.21 ± 0.52 against 2.12 ± 0.31 m, and 28.5 ± 9.9 against 25.6 ± 6.0 degrees), but sun elevation is not: 64.4 ± 7.5 degrees for non-fire against 55.0 ± 15.9 for wildfire, with a fire mode near 35° that has no counterpart in training. Low sun means long shadows and a different appearance for the same structure.

Limitation: confounded acquisition geometry. These spreads overstate their precision, since every scene within an event shares one acquisition. The effective sample size is 19 events, not 11,034 scenes, and acquisition geometry is fully confounded with event identity: wildfire and its captures cannot be separated on this data. Part of the transfer gap reported below may therefore reflect shadow length rather than fire damage, and we cannot bound how much.

3.5 What the classes look like

Figure 4 contrasts the two domains directly. In the non-fire domain the post-disaster image usually retains the structure with altered geometry: torn roofs, debris, standing water. Wildfire damage instead collapses a building to a uniform grey scar with the footprint faintly visible, while undamaged buildings often sit in surroundings that changed completely (scorched vegetation) with the roof untouched.

That last observation generates a falsifiable prediction, made before any model was trained: a change-detection model learns "the scene changed" ⇒ "the building is damaged", so under wildfire it should flag undamaged buildings as damaged. Section 6 reports the outcome.

4 Data cleaning

Nothing is removed silently. Every rule states its threshold, its cost and its per-split removal rate, so a rule that quietly favours one split can be caught. Table 2 is the ledger.

The two rules that cost data are the un-classified subtype (3.29% of all polygons) and sub-8-pixel footprints (3.51% of the labelled remainder). Both were checked per split. The un-classified rate is 3.67 / 4.08 / 2.09 / 3.30% across train / val / test_id / test_ood, and the tiny-footprint rate is 3.77 / 3.30 / 2.99 / 3.49%. Neither is concentrated in `test_ood`, so neither flatters the transfer result.

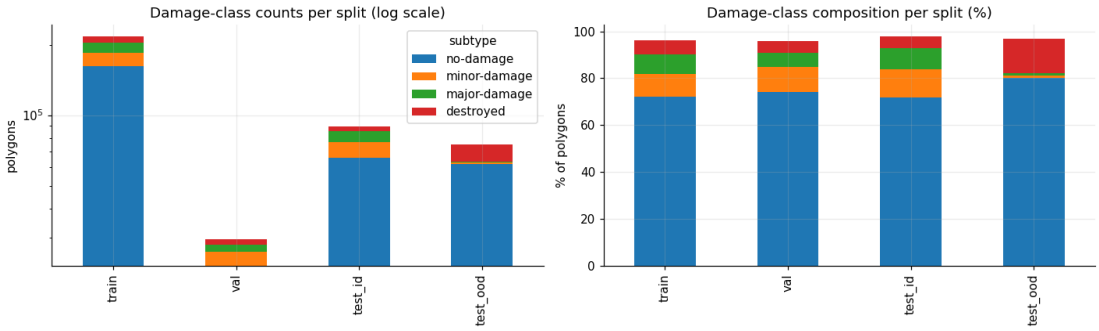


Figure 1: Damage-class distribution per split. Left: counts on a log scale. Right: composition in per cent. The `test_ood` bar is qualitatively different from the other three, with the intermediate classes almost vanishing and destroyed more than doubling.

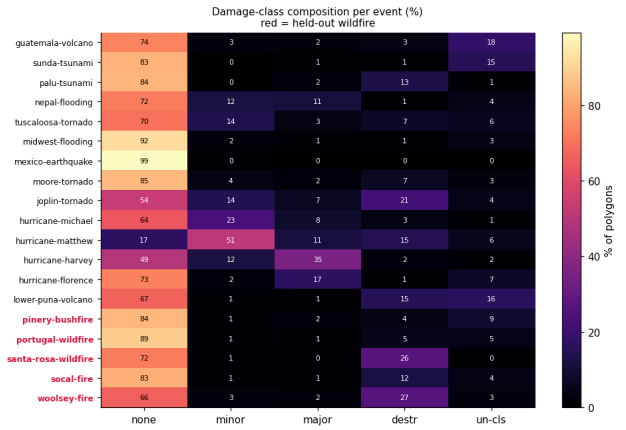


Figure 2: Damage-class composition per event, in per cent of that event's polygons. Held-out wildfire events in red. The fire rows share a signature that no non-fire event reproduces: near-zero minor/major with elevated destroyed.

Table 2: Cleaning ledger. Removal is reported against the scope each rule operates on. The occlusion scenes and the 27 near-duplicate pairs were flagged for inspection, not deleted.

Rule	Scope	Removed
Missing / unpaired	11,034 scenes	0
Corrupted images	788 files	0
Unreadable labels	11,034 scenes	0
Smoke / cloud	494 scenes	0
un-classified	425,368 polys	14,011
Degenerate polygons	425,368 polys	0
Side < 8 px	411,357 polys	14,458
Near-duplicates	788 scenes	0
Usable after cleaning (93.3% of 425,368)		**396,899**

4.1 The occlusion question: a negative result

An open data-quality question for this project was whether images obscured by smoke should be removed. xBD carries no occlusion label: the metadata block contains only capture geometry. The closest available proxy is the un-classified subtype, which annotators used when a building could not be assessed.

The occlusion proxy (`un-classified`) was tested against four whole-tile statistics chosen because smoke and thin cloud are typically bright, low-contrast, low-saturation and edge-poor, on a 788-tile sample enriched toward high-un-classified scenes, restricted to the 494 tiles carrying at least ten assessed buildings. Only edge density showed any correlation at all (Spearman -0.183, $\sim$3% of rank variance); brightness, saturation and contrast were negligible. Flagging the top 2% of a composite score returned only `sunda-tsunami` and `lower-puna-volcano` scenes, none from either test split: no signal justified a threshold, and none of the flagged scenes touched a reported metric.

Table 3: Extracted patch pairs per split and class. Note the support for minor and major in `test_ood`: every per-class number we report for those two classes rests on these counts.

	none	minor	major	destr	**total**
`train`	55,284	7,661	6,265	3,706	72,916
`val`	15,701	2,571	1,164	1,010	20,446
`test_id`	20,500	2,840	2,512	1,421	27,273
`test_ood`	12,056	**116**	**138**	2,317	14,627
Total					**135,262**

4.2 Leakage checks

A perceptual-hash duplicate test over the 788-scene quality-control sample returned 27 near-duplicate pairs, 8 crossing a split boundary. Every cross-split pair traces to a handful of low-texture tiles (open water, unbroken forest, solid cloud) colliding across unrelated events hundreds of kilometres apart, not plausible as genuine overlap. The cause is hash degeneracy: a 64-bit difference hash of a low-texture tile collapses toward a constant, so flat tiles collide regardless of content. We find no evidence of leakage, consistent with the scene-level split design.

5 Method

5.1 Patch construction

One sample is two $64 \times 64 \times 3$ `uint8` patches cropped from the same pixel bounding box in the pre- and post-disaster tile, plus one integer label. Since the tiles are already co-registered, no warping is needed; preprocessing is crop, resize, scale to $[0, 1]$ and ImageNet normalisation. Augmentation (flips, 90-degree rotations, brightness jitter) applies identically to both images of a pair, since independent jitter per branch would manufacture a difference where none exists, precisely the signal the model is meant to measure.

Scenes were sampled under a compute budget, stratified by event and restricted to scenes with at least five usable buildings, yielding 135,262 patch pairs from 1,698 scenes (Table 3). Patches are written to memory-mapped arrays rather than individual image files, which keeps the working set on disk and the loader fast.

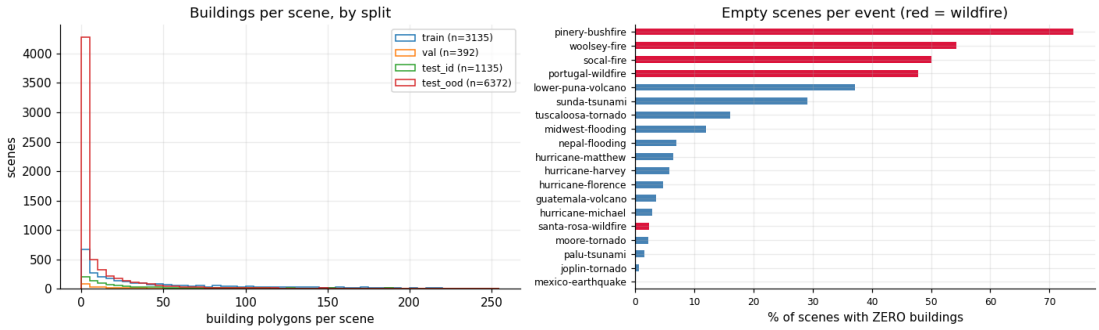
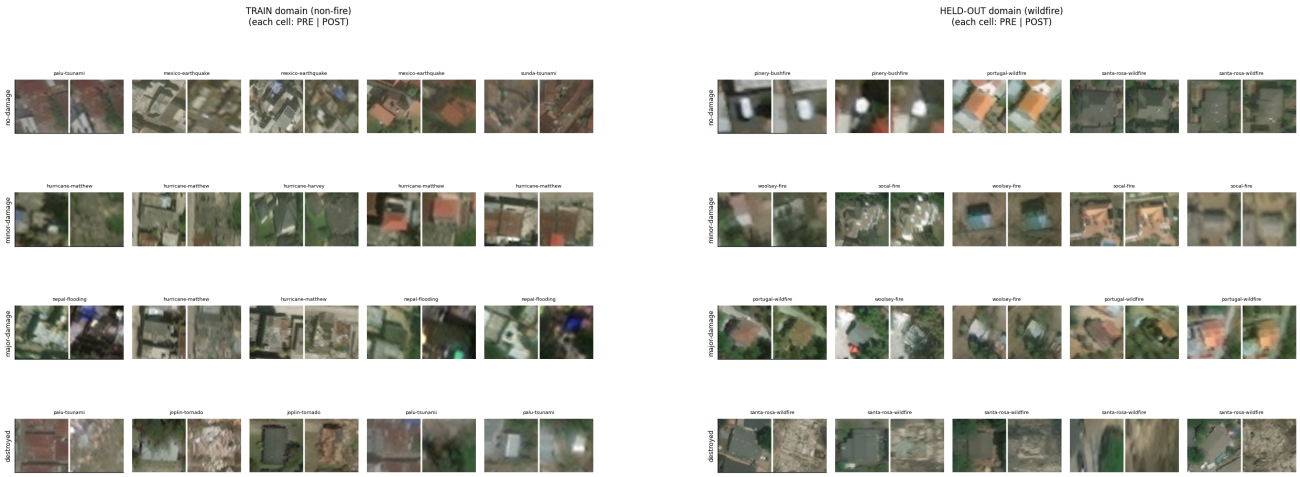


Figure 3: Left: distribution of building count per scene by split. Right: share of scenes with zero buildings, per event (wildfire in red). Four of the five wildfire events are close to or above half empty; pooled over all wildfire scenes the figure is 54.1%.



(a) Training domain (non-fire). **(b)** Held-out domain (wildfire).

Figure 4: Representative building patches, one row per damage class. Each cell shows the pre-disaster crop on the left and the post-disaster crop on the right.

5.2 Architecture

The task is a comparison, not classification of a single image. Using only the post-disaster patch discards the reference, while stacking pre and post into a six-channel input forces the first convolution to be relearned, discarding the pretrained weights exactly where they transfer best. Both models therefore use a siamese encoder with shared weights and a small head reading

$$\left[\, f_{\text{pre}},\;\; f_{\text{post}},\;\; |f_{\text{pre}} - f_{\text{post}}| \,\right].$$

The absolute difference is the explicit change signal; keeping the raw features too lets the head use appearance as well. Shared weights halve the parameter count and force both branches into a common feature space, which is what makes the difference term meaningful.

Model A (pretrained). Uses an ImageNet-pretrained ResNet-18 [3, 4] encoder, 11.57M parameters in total. At a 64×64 input, depth stops paying: architectures designed for 224 px inputs downsample away the little spatial resolution available, and a vision transformer would reduce a 64 px image to a handful of tokens without the data to compensate for the missing convolutional prior. ResNet-18 also keeps the comparison honest, since a much larger backbone would make any win attributable to size rather than pretraining.

Model B (from scratch). Replaces that encoder with a compact stack built for this problem: four blocks of two 3×3 convolutions with BatchNorm and ReLU followed by 2×2 max-pooling, at channel widths $32 \to 64 \to 128 \to 256$, then global average pooling. Total 1.37M parameters. Everything downstream of the encoder is identical.

5.3 Training

Model A follows a linear-probe-then-fine-tune schedule [5]: phase 1 trains only the head on a frozen backbone for 5 epochs at 10^{-3}, so the randomly initialised head does not push large gradients into good weights; phase 2 unfreezes `layer3` and `layer4` for up to 45 further epochs at 10^{-4}, since early layers transfer essentially unchanged while the late layers must be repurposed. Model B has no pretrained weights to protect, so it trains from step one at 10^{-3} over the same 50-epoch budget.

Both use AdamW with cosine annealing, weight decay 10^{-4}, weighted cross-entropy with the class weights above, batch size 128, mixed precision, seed 42, early stopping on validation macro-F1 with patience 8, and checkpointing on the same criterion. Both are implemented in PyTorch [7] and ran locally on an NVIDIA GeForce RTX 4070 Laptop GPU; Model A stopped after 43 epochs (best checkpoint epoch 35); Model B stopped after 16 epochs (best checkpoint epoch 8), in 14.2 minutes at 53 s per epoch.

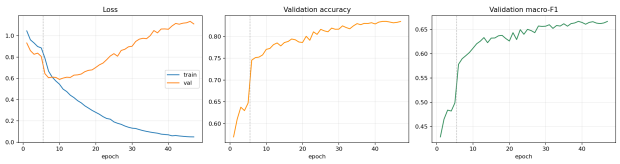


Figure 5: Model A training history. Validation loss bottoms out at epoch 11 and then rises steadily, while validation macro-F1 keeps improving to epoch 35.

Table 4: Headline results. Two trained models, three evaluation sets each. P is the pretrained ResNet-18 encoder, S the from-scratch encoder. `val` and `test_id` are seen disaster types, `test_ood` is the held-out wildfire set; the drops are `test_id` to `test_ood`.

Split		Acc.	Macro-F1	AUC
`val`	P	**0.830**	**0.666**	**0.910**
	S	0.763	0.597	0.902
`test_id`	P	**0.808**	**0.650**	**0.908**
	S	0.768	0.615	0.903
`test_ood`	P	**0.774**	**0.429**	**0.814**
	S	0.630	0.377	0.790
Abs. drop	P	−0.034	−0.221	−0.094
	S	−0.138	−0.238	−0.113
Rel. drop	P	−4.2%	−33.9%	−10.4%
	S	−17.9%	−38.7%	−12.5%

Two observations from the curves (Fig. 5) are worth noting. Unfreezing lifted Model A's validation macro-F1 from 0.499 to 0.580 in a single epoch, the clearest evidence that a frozen ImageNet backbone is insufficient for satellite imagery. From its epoch-11 minimum (0.588), validation loss then climbs to 1.027 by the best checkpoint and 1.130 by the end, while macro-F1 keeps improving (0.633 to 0.666). This is expected under a weighted loss on imbalanced data, since the model grows over-confident faster than it grows wrong, and it is why loss would have been the wrong early-stopping criterion. Model B decouples as well, but in the opposite direction: over its much shorter run its validation loss is still falling at the final epoch (0.919 to 0.617) while macro-F1 peaks at epoch 8 (0.597) and never recovers. In neither model does validation loss track the metric the task is scored on.

6 Results

Table 4 gives the headline numbers. For both models `val` and `test_id` agree closely (0.666 against 0.650 macro-F1 for P, 0.597 against 0.615 for S); this agreement indicates that the in-distribution estimate is not an artefact of a favourable validation split, so the `test_ood` gap reported below reflects a genuine domain effect rather than sampling noise.

6.1 Per-class behaviour

The aggregate drop is misleading, and the per-class view (Fig. 6) shows why.

For the pretrained model, destroyed scores F1 0.758 on wildfire against 0.687 in domain, at precision 0.872, and no-damage reaches precision 0.966. Part of that F1 gain is a base-rate effect, since destroyed is three times more prevalent out of domain (15.8% against 5.2%) and precision rises with prevalence; recall in fact falls, 0.749 to 0.670. Recognition of destroyed structures nonetheless survives a shift to imagery the model never trained on. The degradation is confined entirely to the two intermediate classes, whose F1 falls from 0.480 and 0.529 in domain to 0.038 and 0.043 on wildfire. What breaks is precision, down to 0.020 and 0.024; recall does not vanish (0.284 and 0.203).

6.2 Error structure

The over-prediction is systematic rather than isolated. Against 116 true minor and 138 true major patches, the pretrained model emits 2,814 intermediate predictions and the from-scratch model 4,713, an over-prediction of 11.1× and 18.6×. That alone caps combined precision at $254/2{,}814 = 0.090$ whatever the features look like; the measured 0.020 and 0.024 sit a further factor of four below that ceiling. Figure 7 shows the pretrained model's failure mode alongside a correct prediction from each test split.

The direction of the errors is equally consistent. For the from-scratch model, whose full confusion matrices are available (Fig. 8), 81.1% of out-of-domain errors over-estimate severity against 18.9% that underestimate, and 35.9% of truly undamaged wildfire buildings (4,324 of 12,056) are flagged as damaged. The pretrained model shows the same failure mode less severely: its no-damage recall of 0.805 on `test_ood` implies a 19.5% false-alarm rate. Both confirm the prediction made before training.

6.3 Pretrained against from-scratch

This section addresses the project's second research question directly. Both models train on identical patches, splits, class weights and evaluation code, differing only in the encoder, isolating the effect of ImageNet pretraining from every other design choice. Pretraining outperforms the from-scratch encoder on every split and metric in Table 4, and the advantage is largest where it matters most: on `test_ood` it adds +0.143 accuracy, +0.052 macro-F1 (a 13.9% relative gain over the from-scratch model's 0.377) and +0.024 ROC-AUC. The relative macro-F1 drop from `test_id` also shrinks, from 38.7% to 33.9%, so pretraining not only raises the floor but makes the domain-shift degradation proportionally smaller.

Per class, the clearest difference is destroyed: 0.758 against 0.662 out of domain, 0.687 against 0.563 in domain. The intermediate classes are not meaningfully distinguishable between the models, since both are far below usable precision, and at $n = 116$ and $n = 138$ the small reversals fall within noise. One caveat: this is not a pure pretraining ablation, since the pretrained encoder is also 8.4× larger. A third run with a randomly initialised ResNet-18 would separate pretraining from capacity; we have not run it (Section 9).

7 Discussion

Evidence for a calibration failure rather than a representation failure. Four observations support this reading of the 33.9% macro-F1 drop.

(1) destroyed scores higher out of domain than in it (0.758 against 0.687, part of which is its threefold higher prevalence there), its recall holds at 0.670, and no-damage precision is 0.966: the two classes

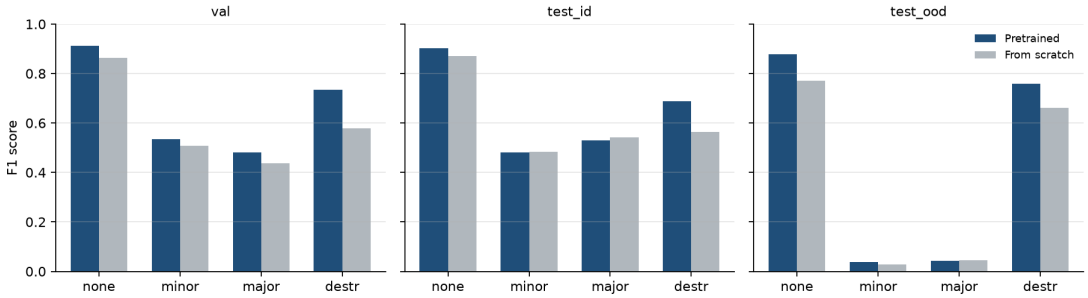


Figure 6: Per-class F1 for both models on the three evaluation sets. In domain the two are close on the intermediate classes and pretraining mostly buys destroyed; out of domain both collapse on minor and major, and the pretrained advantage on destroyed widens.

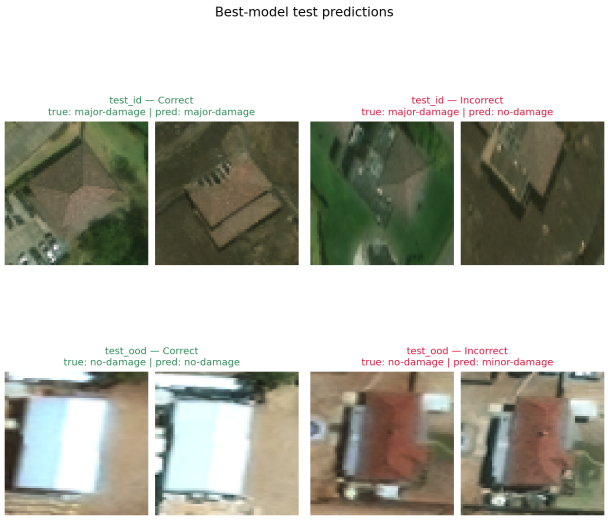


Figure 7: Pretrained model: one correct and one incorrect prediction on `test_id` and on `test_ood`, pre-disaster crop left and post-disaster crop right in each pair. The `test_ood` error is a truly undamaged building predicted minor-damage, the over-flagging failure mode quantified above.

that actually occur under wildfire are recognised well.

(2) The errors are directionally consistent, 81.1% overestimating severity for the model whose full confusion matrix is available. A consistent bias is the signature of a shifted prior; a representation failure would instead produce diffuse, symmetric confusion.

(3) ROC-AUC, which does not depend on a decision threshold, holds at 0.814 against 0.908 in domain: the ranking of predictions by severity survives, and it is the decision boundary that fails.

(4) The collapse is largely forced by the class-frequency shift. minor and major make up 10.5% and 8.6% of training patches but only 0.79% and 0.94% of wildfire patches. The pretrained model's 2,814 intermediate predictions against 254 true cases cap combined precision at 0.090 whatever the features look like, four times the 0.020 and 0.024 measured. The prior shift therefore accounts for most of the drop but not the last factor of four, which is a genuine loss of discrimination on the two classes wildfire almost never produces.

This distinction has direct operational consequences. A representation failure would mean the approach cannot be deployed on a new disaster type at all. A calibration failure means the ranking of predictions is usable and only the decision rule needs adjusting, a more tractable problem with an established remedy in prior correction [6], which rescales class posteriors by the ratio of target to source priors without retraining. A model of this kind deployed on an unlabelled new disaster type would reliably separate destroyed from intact structures while substantially over-reporting intermediate damage. For triage, i.e. prioritising where responders look first, that failure mode is comparatively benign; for damage quantification, and especially aid allocation, it is not.

8 Impact and recovery analysis

8.1 Data and method

To ask whether damage falls unequally across communities, we take the per-building WGS84 centroids in the xBD labels, restrict to the ten US events with available census geography, and spatially join them to 2020 TIGER census block groups. Each building contributes an ordinal damage score (no-damage = 0 to destroyed = 3); the unit of analysis is the block-group mean, dropping groups with fewer than 20 buildings. The join reaches 185,045 buildings; the building threshold and the ACS merge leave 308 block groups, joined to American Community Survey covariates for 2017 [8].

Two data-quality problems were handled first, both reported rather than absorbed silently.

Census sentinel values. The ACS extract encodes missing values as −666666666 rather than nulls; 17 of the 308 block groups carry one in median income, home value or year built, which would make those coefficients meaningless if left in place. We drop them, leaving 291 block groups holding 108,491 of the joined buildings.

Two unusable columns. The poverty rate is empty for every block group, and the hazard-exposure proxy is identically zero since no fire-perimeter, storm-track or coastline layer was ever joined. Both are dropped rather than reported as null effects.

The exposure control is therefore the built environment only (building density and median building age),

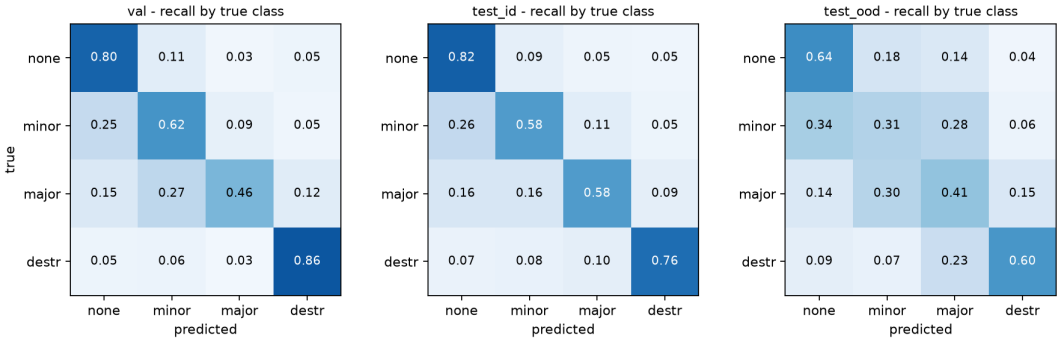


Figure 8: Confusion matrices for the from-scratch model, normalised per true class (recall). The `test_ood` panel shows probability mass pushed off the diagonal in one direction, toward greater severity, rather than scattered.

Table 5: Standardised OLS coefficients on block-group mean damage score, $n = 291$. Each column adds controls to the one before it: socioeconomic only, plus built environment, plus event fixed effects.

	naive	**+ built**	**+ FE**
Median income	$+0.211^{**}$	$+0.225^{***}$	$+0.057$
Median home value	-0.150^{**}	-0.139^{*}	-0.039
Renter share	$+0.015$	$+0.032$	-0.016
Building density		-0.085^{*}	-0.103^{**}
Building age		-0.102^{*}	-0.096^{*}
R^2	0.045	0.072	0.225
adjusted R^2	0.035	0.055	0.186

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

plus, in the strongest specification, event fixed effects—the sharpest exposure control this data supports, since comparing two block groups inside the same disaster holds the hazard roughly constant.

8.2 Results

Table 5 summarises the result. Regressed on socioeconomic variables alone, median income carries a significant positive coefficient ($+0.211$, $p = 0.002$), strengthening further with built-environment controls ($+0.225$, $p = 0.0009$)—taken at face value, evidence that richer block groups suffered more damage.

Adding event fixed effects destroys the effect: the income coefficient falls to $+0.057$, no longer distinguishable from zero ($p = 0.44$), while R^2 more than triples, from 0.072 to 0.225. Almost all the apparent socioeconomic signal was between-event variation: `hurricane-harvey` block groups have both high income ($98k median) and high damage (mean score 0.88), while `santa-rosa-wildfire` and `moore-tornado` have moderate income and low damage.

Figure 9 shows the same thing without a model. Pooled across events, the relationship is U-shaped, not monotone: mean damage score is 0.58 in the poorest income quintile, falls to 0.33 in the middle, and rises to 0.61 in the richest—a mixture of two exposure stories (low-income tracts in tornado paths, high-income tracts in coastal and wildland-urban-interface zones), not a single inequality gradient. Computing income terciles within each event, over the six events with at least 15 block groups, flattens it to 0.51 / 0.41 / 0.51, and the per-event direction is inconsistent in sign: damage rises with income across `hurricane-harvey` (0.77 to 1.01) and falls with it across `joplin-tornado` (0.90 to 0.41).

What survives event fixed effects is the built environment: building density (-0.103, $p = 0.006$) and median building age (-0.096, $p = 0.012$) remain significant and barely move (Fig. 10). Denser, older block groups record less damage on average—against the usual intuition that older housing is more vulnerable—most plausibly a settlement-pattern effect: dense, older cores sit away from the coastal and interface edges where these hazards concentrate.

8.3 Pricing damage, and checking the price against real records

An ordinal damage class says nothing about the size of the loss, and the project specification asks about recovery as well as damage. Every building's damage class is therefore converted to a dollar figure using the HAZUS-MH per-damage-state loss ratios [10] (0% for no-damage, 5% minor-damage, 30% major-damage, 100% destroyed) applied to its block group's median home value. The rule is not disaster-specific: a destroyed house costs its replacement value whichever hazard destroyed it. Over the 108,491 buildings in the 291 analysis block groups this prices $6.76B of modelled loss, wildfire alone accounting for 70% (Californian home values are high and fire tends to destroy outright), hurricane 22%, tornado 7%, and the single flood event under 1%, so the dollar layer is not simply a restatement of flood risk.

That price was checked against three independent, real-world records rather than taken on faith. Against 6,301 NFIP settlements [11] inside the same block groups, the modelled price lands at 0.32–0.85× the adjuster's on-site assessment: the right order of magnitude, and consistently conservative. Against 9,302 CAL FIRE post-fire structure inspections [12] for the three wildfire events, inspectors measured an average loss of 91.4% where the assumed ratios give 92.7%, within 1.3 points. And against 133,240 FEMA household loss verifications [11] spanning all ten events, the modelled price runs well above FEMA's figure, as expected: FEMA's verified loss records the damage the aid programme will pay for, a floor on rebuild cost rather than an estimate of it.

8.4 Recovery capacity is where the real inequality is

Figure 11 makes a point the regression above cannot. Physical damage is close to flat across income (10.6% of building value lost in the poorest quintile against

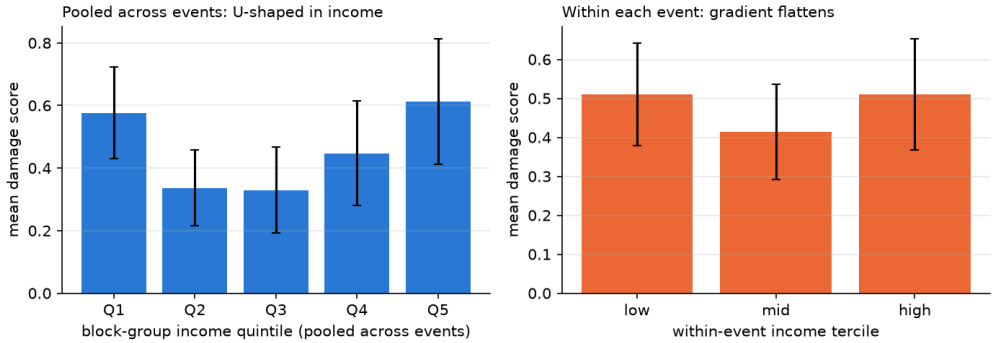


Figure 9: Left: pooling all ten events, mean damage score is U-shaped in income, highest in the poorest and the richest quintiles and lowest in the middle. Right: once income terciles are computed *within* each event, the gradient flattens and the 95% intervals overlap completely.

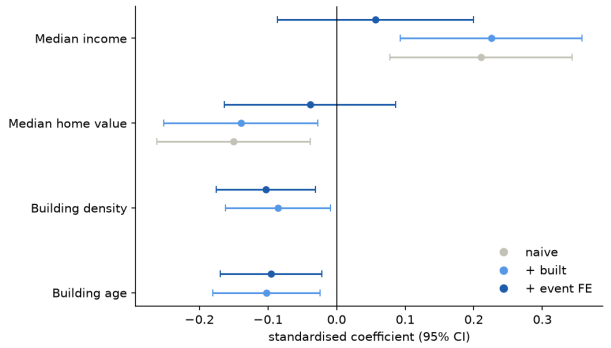


Figure 10: Standardised coefficients with 95% intervals across the three specifications. The two socioeconomic effects shrink toward zero when event fixed effects enter; the two built-environment effects do not.

11.7% in the richest, non-monotone), but the same damage map priced in dollars is steep, rising from \$13k to \$85k per building, a $\sim$7$\times$ gradient, simply because a home in a richer block group costs more to begin with.

Who actually pays that bill tracks income even more sharply (Fig. 12). Restricting to the four water events, where NFIP publishes a claim for every settled property, claims filed per damaged building rise steeply with income (Spearman $\rho = 0.63$): in the poorest income quartile the median block group files for 5% of its damaged buildings against 51% in the richest, and in aggregate the two quartiles file for 7% and 48%. Splitting the repair bill three ways tells the same story: insurance covers 24% of the poorest quartile's modelled loss and FEMA grants another 16%, leaving 60% covered by nothing, while in the richest quartile insurance alone covers 66% and only 33% goes uncovered.

FEMA's own applicant records extend this picture to every disaster type [11]. Among 115,192 aid applicants who reported a household income, 99% of them owner-occupants, the insured share rises from 38% in the poorest household-income band to 95% in the \$120–175k band and holds there above it, and the gradient holds inside every disaster type. This also corrects a flood-only impression: wildfire households are the best insured of the four (75%), flood households the worst (60%), since a standard homeowner policy covers fire and wind while flood needs a separate NFIP policy.

8.5 Auditing the predicted damage map

Every result so far uses xBD ground-truth labels, by design (Section 8.6 explains why). The last question this data can answer directly is what changes if a relief programme used the Part 2 models' predictions instead of verified labels. Joining the two models' output on their held-out splits to the same census geography gives 12,438 buildings across seven events (three hurricanes, one flood, three wildfires) and 122 block groups; tornado is absent for an unrelated reason, since all three tornado events sit entirely inside the Part 2 training split and have no held-out predicted map.

We split a fixed \$1B relief budget by income tercile twice, once on the true damage map and once on the pretrained model's predicted map, under two allocation rules (equal payment per damaged building, and payment proportional to modelled dollar loss), and measure how many dollars land in a different income tercile than the labels would send them. On the seen disaster types the predicted map misdirects \$30–78M of every \$1B; on wildfire, never seen in training, it is \$112–200M (Fig. 13), and the direction is systematic: consistent with the intermediate-class over-prediction in Section 6, the model invents damage in mid-income block groups and withdraws funding from the high-income block groups where wildfires actually destroyed housing. The predicted map also over-states total dollar loss on this sample by 19%—not a flaw specific to the pretrained model, since the from-scratch model's audit numbers are of the same order (\$37–81M seen, \$137–179M wildfire).

8.6 What this means for deployment

Three cautions follow, and they compound. First, the naive specification would have supported a confident but incorrect policy claim: a pipeline that regressed damage on income and reported the coefficient would publish a significant positive income effect that does not survive the most basic exposure control available. The confounder here is not subtle; it is simply which disaster struck which place.

Second, the recovery-capacity analysis in Section 8.4 shows an inequality the damage regression cannot see: physical damage is close to flat across income, but dollar loss and the capacity to cover it are not. A relief

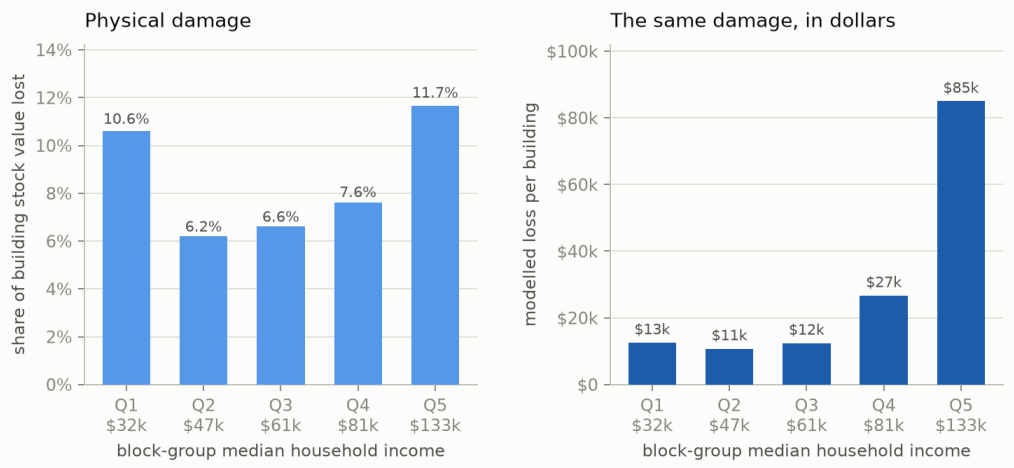


Figure 11: Left: share of building value lost, by income quintile. Right: the same damage priced in dollars per building. Physical damage is flat; the dollar figure is not.

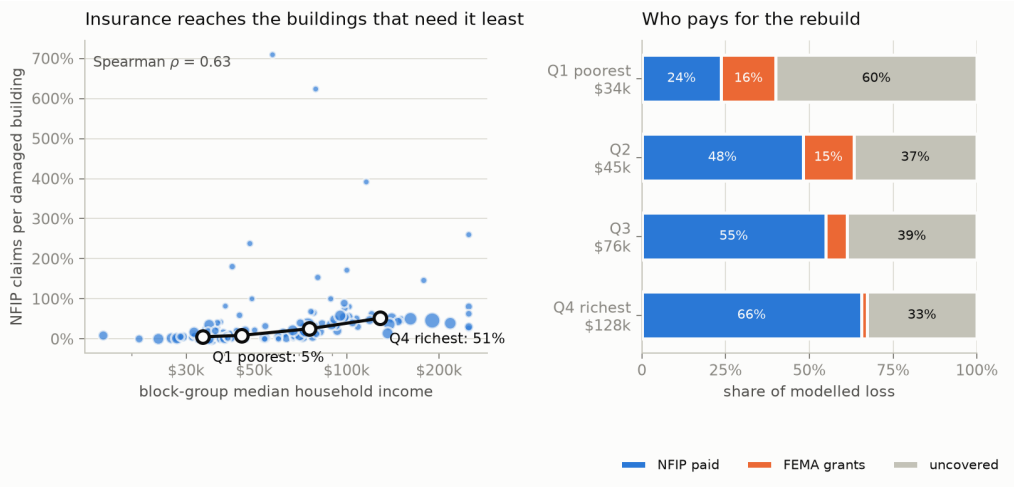


Figure 12: Left: NFIP claims filed per damaged building against income, four water events. Right: the repair bill split among insurance, FEMA and nobody, by income quartile.

rule keyed to dollar damage, or to insurance status, is regressive before any model is involved.

Third, Section 6 showed the model's errors are not random: it over-flags intermediate damage at a rate that depends on disaster type, and Section 8.5 shows what that costs in practice, moving real relief money along income lines, more so on the disaster type the model has never seen. Rapid automated damage maps are useful for triage; using them as evidence for aid allocation without verified labels risks a specific, income-correlated error, in the direction the model already over-predicts.

9 Limitations

- **Thin out-of-domain support.** minor and major rest on 116 and 138 true examples in `test_ood`. These two classes drive the macro-F1 story, so their per-class figures carry wide uncertainty.
- **Annotator selection.** Every number is measured on buildings annotators were able to assess, since un-classified polygons are excluded, so true zero-shot difficulty is somewhat higher than reported.
- **Partial checks and sampling.** The duplicate test covers 788 of 11,034 scenes, and model results use a stratified sample of 1,698 scenes (the acquisition-geometry confound is discussed in Section 3.4).
- **Not a pure pretraining ablation**, since the encoders differ in initialisation and in size (11.57M against 1.37M parameters); and model selection uses `val`, which is in-distribution, because selecting on the out-of-domain set would leak the test set.
- **Ecological inference in Part 3.** Block-group averages cannot support claims about individual households, no true hazard-exposure layer was available, and ten events is a small basis for fixed effects.
- **Recovery-analysis assumptions.** One median home value stands in for every building in a block group, and four HAZUS loss ratios stand in for a continuous damage distribution. Tornado has no inspection dataset comparable to CAL FIRE's, so its loss ratios rest on the HAZUS assumption alone. Insurance take-up from FEMA records describes owner-occupants who applied for aid, not the general population, and the predicted-damage audit covers seven of the ten events because the three tornado events never appear in a held-out split.

10 Conclusions and Future Research

This project set out to answer two questions: whether building-damage classification transfers to a disaster

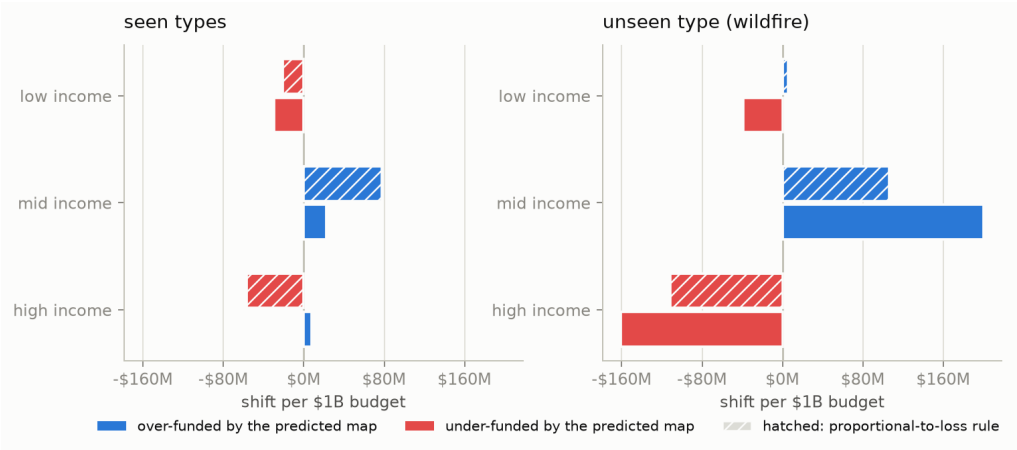


Figure 13: Dollars per $1B misdirected across income terciles, predicted versus true damage map (pretrained model). Blue over-funds, red under-funds. The wildfire panel is larger and directionally consistent; the seen-type panel is not.

type withheld from training, and whether ImageNet pretraining improves that transfer over a comparable from-scratch encoder. On the first, a siamese change-detection CNN trained on 14 non-fire disasters and evaluated zero-shot on five wildfire events loses a third of its macro-F1, but the loss is confined to the two damage classes that barely exist under wildfire, while destroyed-versus-intact discrimination improves out of domain. On the second, ImageNet pretraining outperforms the from-scratch encoder on every split and metric, with the largest margin on the wildfire set. A supplementary impact analysis finds that raw damage is not distributed along an income gradient once the disaster event is controlled for, but that does not extend to recovery: priced in dollars and measured against real insurance and federal-aid records, the same damage falls on households with sharply unequal capacity to pay for it, and auditing the Part 2 models' own predictions shows an automated relief rule would move real money along income lines, more so on the disaster type the models never trained on. Outstanding work, in priority order:

1. **Test the calibration hypothesis directly.** Apply prior correction [6] to the out-of-domain predictions of both models and measure how much of the macro-F1 gap closes without retraining. If most of it closes, that is the strongest result the project can report.
2. **Few-shot ablation**: add 1 to 5% of wildfire scenes to training and re-measure the gap.
3. **Randomly initialised ResNet-18** as a third run, to separate pretraining from capacity.
4. **Real hazard-exposure layers** for Part 3, in place of the event-fixed-effects proxy, and a tornado inspection dataset comparable to CAL FIRE's to remove the last HAZUS-only loss ratio.

Reproducibility. Every figure, table and number in this report is generated from a CSV artefact written by the notebooks and scripts in the repository; nothing is hand-computed. Seed 42 throughout. Environment setup, data sources, and a map from each notebook to the results and figures it produces are documented in the repository `README.md`.

References

[1] R. Gupta *et al.*, "xBD: A Dataset for Assessing Building Damage from Satellite Imagery," *arXiv:1911.09296*, 2019.

[2] E. Weber and H. Kané, "Building Disaster Damage Assessment in Satellite Imagery with Multi-Temporal Fusion," *arXiv:2004.05525*, 2020.

[3] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.

[4] J. Deng *et al.*, "ImageNet: A Large-Scale Hierarchical Image Database," *CVPR*, 2009.

[5] A. Kumar, A. Raghunathan, R. Jones, T. Ma and P. Liang, "Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution," *ICLR*, 2022.

[6] M. Saerens, P. Latinne and C. Decaestecker, "Adjusting the Outputs of a Classifier to New a Priori Probabilities," *Neural Computation*, 14(1):21-41, 2002.

[7] A. Paszke *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *NeurIPS*, 2019.

[8] U.S. Census Bureau, American Community Survey 5-Year Estimates (2017) and TIGER/Line block-group boundaries (2020).

[9] xView2 Challenge, `https://xview2.org`.

[10] Federal Emergency Management Agency, *Hazus – Multi-Hazard Loss Estimation Methodology (Hazus-MH)*, Technical Manual, `https://www.fema.gov/flood-maps/tools-resources/flood-map-products/hazus`.

[11] Federal Emergency Management Agency, OpenFEMA API: *FIMA NFIP Redacted Claims*, *Individuals and Households Program Valid Registrations*, and disaster declaration summaries, `https://www.fema.gov/about/openfema/data-sets`.

[12] California Department of Forestry and Fire Protection (CAL FIRE), *Damage Inspection (DINS) Data*, `https://www.fire.ca.gov/programs/fire-prevention/wildfire-preparedness/damage-inspection-dins-data`.